GEOMETRÍA EN
MOVIMIENTO

1. Funciones Base y Lineales

1. Constante horizontal `y = a`	**2.** Constante vertical `x = a`
3. Identidad `y = x`	**4.** Función Lineal `y = a*x + b`
5. Valor Absoluto `y = abs(x)`	**6.** Función Signo `y = sgn(x)`
7. Escalón unitario	**8.** Parte entera (Floor)

`y^2 = x^2 / (1-x^2)`

`x^2*(x^2 + y^2) = a^2*y^2`

29. Estrofoide
`y^2 = x^2*(a+x)/(a-x)`

30. Bifolio
`(x^2 + y^2)^2 = ax^2y`

4. Secciones Cónicas (Implícitas)

31. Círculo
`x^2 + y^2 = r^2`

32. Elipse
`(x/a)^2 + (y/b)^2 = 1`

33. Hipérbola

34. Hipérbola equilátera

5. Trigonometría y Derivadas

6. Coordenadas Polares y Otras

`r = a*exp(b*theta)`	`r = sqrt(theta)`
55. Espiral Lituus `r = 1/sqrt(theta)`	**56.** Cardioide `r = a*(1 - cos(theta))`
57. Limacon (Caracol) `r = b + a*cos(theta)`	**58.** Rosa de 3 pétalos `r = cos(3*theta)`
59. Rosa de 4 pétalos `r = cos(2*theta)`	**60.** Rosa de n pétalos `r = cos(k*theta)`
61. Lemniscata polar `r^2 = cos(2*theta)`	**62.** Trifolium `r = cos(theta) * sin(theta)^2`
63. Nefroide polar `r = a*(1 + 2*cos(theta))`	**64.** Hipopede `r^2 = 4b(a - b*sin(theta)^2)`

9. Superficies Implícitas 3D

85. Esfera
`x^2 + y^2 + z^2 = r^2`

86. Elipsoide
`(x/a)^2 + (y/b)^2 + (z/c)^2 = 1`

87. Hiperboloide de 1 hoja
`x^2 + y^2 - z^2 = 1`

88. Hiperboloide de 2 hojas
`x^2 + y^2 - z^2 = -1`

89. Toroide (Dona)
`(sqrt(x^2+y^2)-R)^2 + z^2 = r^2`

90. Cubo redondeado
`x^4 + y^4 + z^4 = 1`

91. Superficie de Barth

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12. Arte Matemático y Formas Especiales

103. Cinta de Moebius

```
x=(1+v/2*cos(u/2))cos(u),
y=(1+v/2*cos(u/2))sin(u),
z=v/2*sin(u/2)
```

104. Distribución de Cauchy

```
y = 1 / (pi * (1 + x^2))
```

105. Botella de Klein

```
x=(2+cos(u/2)*sin(v)-sin(u/
2)*sin(2v))*cos(u), ...
```

106. Corazón 3D (Implícito)

```
(x^2 + (9/4)y^2 + z^2 - 1)^3 - x^2*z^3
- (9/80)y^2*z^3 = 0
```

107. Swirl (Remolino 3D)

```
z = sin(sqrt(x^2+y^2) - atan(y/x))
```

108. Distribución Log-Normal

```
y = (1/(x*sqrt(2*pi))) * exp(-
0.5*(log(x))^2)
```

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`r = 1 + 2*sin(theta/2)`	`r = sin(theta) * cos(10*theta)`
127. Variante de mariposa polar `r = exp(sin(theta)) - 2*cos(4*theta)`	**128.** Lemniscata polar absoluta `r = sqrt(abs(cos(2*theta)))`
129. Estrella de mar asimétrica `r = 2 + sin(5*theta/2)`	**130.** Espiral oscilante senoidal `r = theta * sin(theta)`

15. Curvas de Lissajous Complejas (x(t), y(t))

131. Lissajous nudo simple	**132.** Lissajous caja cerrada (4:5)
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151. Red de cuadrados bisecados
`cos(x) + cos(y) = 0`

152. Círculos deformados
`sin(x)*cos(y) = 0.5`

153. Hipérbolas anidadas (Cruz)
`sin(x*y) = 0`

154. Hipérbolas con asíntotas
`cos(x*y) = 1`

155. Matriz de óvalos aislados
`sin(x) * sin(y) = 0.1`

156. Interferencia hiperbólica
`sin(x^2 - y^2) = 0`

157. Anillos por tangente
`tan(x^2 + y^2) = 1`

158. Tejido topológico complejo
`x*sin(y) = y*cos(x)`

159. Cuadrícula de rombos deformados
`abs(sin(x)) + abs(sin(y)) = 1`

160. Singularidad de impacto
`sin(10x) + sin(10y) = 10xy`

20. Astrofísica y Relatividad

181. Paraboloide de Flamm
```
z = -1 / sqrt(x^2 + y^2 + 0.1)
```

182. Agujero de Gusano (Einstein-Rosen)
```
x=sqrt(v^2+1)*cos(u),
y=sqrt(v^2+1)*sin(u), z=v
```

183. Loxodromia Esférica
```
x=cos(t)/cosh(0.1t),
y=sin(t)/cosh(0.1t), z=tanh(0.1*t)
```

184. Cuerno de Gabriel
```
x=v*cos(u), y=v*sin(u), z=-1/v
```

185. Cono de Luz Estelar (Minkowski)
```
x=v*cos(u), y=v*sin(u), z=v
```

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```
atan(y/x)))
```

23. Bonus: Mecánica y Superficies de Ingeniería (u, v)

201. Tornillo de Arquímedes

```
x=v*cos(u), y=v*sin(u), z=0.5*u
```

202. Engrane Recto

```
x=(4+0.3cos(12u)^6)*cos(u),
y=(4+0.3cos(12u)^6)*sin(u), z=v
```

203. Engrane Cónico

```
x=v*(4+0.3cos(12u)^6)*cos(u),
y=v*(4+0.3cos(12u)^6)*sin(u), z=v
```

204. Resorte Helicoidal

```
x=(3+0.5cos(v))*cos(u),
y=(3+0.5cos(v))*sin(u)
```